Phytopathogenic Enterobacteria, the Conserved Core, and the Oxygen Response: A Summary

The facultative anaerobes *Dickeya dadantii* and *Pectobacterium atrosepticum* are major plant pathogens (1), capable of infecting over 50 percent of angiosperm genera with soft rot, the maceration of tissue with pectinase enzymes (2). They belong to the Gram-negative family Enterobacteriaceae, and can handle growth in the wide-ranging O_2 concentrations of internal plant tissues (3). Facultative anaerobes adapt to changing atmospheric conditions by altering gene expression via diverse mechanisms that are frequently lineage-specific. Such variation from a clade's conserved core genome is a marker for evolution through horizontal gene transfer or mutation events (4). The transcriptional response to oxygen has been well documented in the closely related animal pathogen *E. coli*, but remains poorly understood in *D. dadantii* and *P. atrosepticum*. The respiratory system of *E. coli* is moderated by the transcription factors Fnr, ArcA, NarL, and NarP, and is highly conserved among Gamma-proteobacteria, though regulatory differences exist between lineages (5). Anoxic or oxygen-poor conditions inside plants protect pathogens from oxygen radicals, trigger a virulent phase of rapid division, and secretion of destructive enzymes occurs at a cell density quorum of 10^7 cfu/g of tissue (6). Investigation of the genes involved in their anaerobic regulons and stimulons has important implications in pest control, as the oxygen response in the virulence factors PelA, D, and E (7) may allow infections to spread out of control and destroy whole crops in and out of storage. Especially devastated by the soft rot Enterobacteria is the potato, which suffers from substantial global loss (8). Success has been reported against microbial invasion in orchids, another economically important host plant, by transferring the ferredoxin-like protein from sweet peppers (9), but a complete picture of the bacterial virulence mechanisms and their regulation offers alternative routes for disease eradication.

To study and compare the O_2-dependent transcriptome in enterobacteria, and evaluate the evolutionary relationships of these regulatory systems, Babujee's research team investigated the effects of differing O_2 concentration among *Dickeya dadantii* 3937 and *Pectobacterium atrosepticum* SCRI1043, and compared the transcriptional profiles in the two to available data for *Escherichia coli* K-12. Three replicates of both *D. dadantii* and *P. atrosepticum* were grown to mid-log phase in minimal media at their respective optimal temperatures of 30˚C and 25˚C. The group used a gas sparging system with defined criteria for experimental aerobic (25% O_2, 70% N_2, and 5% CO_2) and anaerobic (95% N_2 and 5% CO_2) conditions. The differential in O_2 concentrations was expected to induce profound changes in the expression of a diverse collection of genes and/or operons, and the team hoped not to replicate the complex conditions inside a hypothetical plant host, but rather to elucidate the extent of the conserved core within the species' genomes.

An orthology-driven approach utilized a high-density oligonucleotide array to identify molecular relationships, and EBArrays (10) located genes undergoing differential expression under varying conditions. Although the analysis was genome-wide, special attention was given to gene transcripts experiencing greater than a 3-fold change between O_2 levels. Examination of the subsystems allowed for the strongest comparison between species (11), including among whole operons and the often-overlooked sRNA genes. However, OrthoMCL locates protein-coding genes only in searching for orthology, so the sRNA genes were individually compared with BLASTn. Though *Dickeya* and *Pectobacterium* are in a monophyletic clade, the team expected evolutionary events to cause genome divergence among systems, and thus related genes and gene products were grouped if they accomplished a similar biochemical goal, albeit through a differing mechanism. The team discovered many of the relevant sRNA genes were incorrectly annotated in online databases, and the necessary edits were made to reflect the experimental data.

EBArrays revealed high rates of differential expression in both *D. dadantii* (48.5%) and *P. atrosepticum* (13.4%), but the more stringent criterion of a 3-fold change in expression revised these percentages to 9.8% and 7.3% of the two genomes, respectively. The anaerobic stimulon in the two bacteria is a large portion in their genomes, mirroring a finding previously reported for *E. coli*. As the two phytopathogens are ecologically similar and evolutionarily closely related, they were expected to share many patterns of expression changes. Indeed, nearly 3,000 genes in each species were found to exhibit simple 1-1 ortholog relationships. Yet, under the stringent criteria, only 247 *D. dadantii* genes and 196 *P. atrosepticum* genes were predicted to include orthologs in the other bacterium's genome. Out of these groups, less than 100 genes met the stringent criteria for both species, and 81 ortholog sets exhibited similar transcription trends under anaerobic and aerobic conditions. Most of the up-regulated genes are known to be crucial in anaerobic growth and functionality, but somewhat surprisingly, the type VI secretion system was oppositely expressed in the phytopathogens. Most of the genes under differential expression were not orthologous to *E. coli*, and the majority of confirmed orthologs are related to anaerobic metabolism and can be considered part of the conserved core for enterics.

The conserved core levels out just above 2,000 genes across the *D. dadantii*, *P. atrosepticum*, and *E. coli* genomes. As *D. dadantii* and *P. atrosepticum* are more closely related to each other than to *E. coli*, their orthologs exhibited greater consistency in expression pattern; only 39 of the 1-1-1 ortholog groups were congruent in expression, with no fold change criteria. Unexpectedly, the transcriptional regulator *fnr* was much more strongly regulated in *D. dadantii* than in either of the other two examined species. Of the four TCA genes encoding succinate dehydrogenase, only *sdhD* was dramatically down-regulated in *D. dadantii*, while A, B, and C were also down-regulated in *E. coli and P. atrosepticum*. This difference is indicative of a lineage-specific error in the regulatory gene *arcB*. Further comparisons were drawn over the metabolic flexibilities of the organisms, their metal transport systems, pH homeostasis, acid response, and other markers of evolution. Such a small conserved core means that regulation and execution of the anaerobic stimulon is likely diverse in Enterobacteriaceae, even within monophyletic clades. While regulators behave differently between species, they tend to occupy the conserved core, while genes specific to individual species' lineages may alter the stimulon in ways that are not immediately obvious, due to convergent phenotypes from a variety of independently evolved biochemical pathways.

Critique on Dr. Perna's Presentation

Dr. Perna is a cordial and confident professional speaker, and began her presentation with detailed background on her past experience in the *E. coli* Genome Project. She demonstrated how her work with the animal pathogen highlighted the differences between lineage-specific "islands" that may dominate 30 percent of an organism's genome, and what she defined as the conserved core – the smaller collection of genes all members of a lineage of species must have in order to survive. The concise introduction set the tone for a succinct, yet highly informational and organized explanation of her research on the evolution of the oxygen response in enterics.

The presentation was cleanly exhibited on Powerpoint slides, with tactful use of limited important bulleted text. Structurally the talk closely and effectively followed Dr. Perna's work on Babujee and Apodaca's 2012 study of the oxygen response in enteric phytopathogens. Dr. Perna also frequently used the chalkboard to further explain the phylogenetic relationships between *Dickeya*, *Escherichia*, and *Pectobacterium*, and demonstrated how the commonly obtained 100% congruent bootstrap results alone could mislead researchers into constructing inaccurate evolutionary histories of the Enterobacteria. Although the study examined broad and diverse subsystems within the species, including the enzymes fumarase and hydrogenase, the majority of data presented focused on the quantities and types of genetic orthologs found to be differentially expressed. However, Dr. Perna did specify the results of subsystem comparisons, showing that *Dickeya* and *Pectobacterium* use unrelated hydrogenases despite their close relationship, and noting that fumarase functions differently in all three relevant lineages. She made excellent contrasts of magnitude between the 20 orthologs that exhibited a greater than 3-fold difference in expression across all three species, with the over 101,000 genes found to date in Enterobacteria.

When asked about surprisingly low growth rates in *E. coli* K-12, she revealed a potentially problematic weakening in purifying selection, brought about by use in countless controlled lab studies. Dr. Perna did not elaborate on how such unexpected changes in selection may have affected experimental results, but the assumption holds that the organism's conserved core genome should have remained largely intact and unaltered by human influence. Nonetheless, she acknowledged the possibility of obtaining vastly different results if other common E. coli strains were investigated alongside the phytopathogens. A greater challenge in ongoing research is to produce an experimental setup that can address questions in the scope of this study within an environment that adequately models a plant. Dr. Perna explained that a starting point would use the gas sparging system, with gas concentration levels adjusted to match experimentally determined internal plant tissue environments.

Though limited by the cost and pace of full-genome sequencing and subsequent microarrays, data patterns will become stronger with the addition of more annotated genomes from various species of Enterobacteria and their numerous and diverse strains. As only one strain was examined from each of the three enterics, it remains unclear to what extent the anaerobic stimulon and regulon will vary within a single species, or how much each evolutionary event (horizontal gene transfer, loss, mutation, or duplication) affects the interspecies variation. Funding is difficult to obtain for grants requesting massive assays that truly examine a diverse set of individual genomes. These questions will be part of the focus of the group's ongoing future research as more genomes are sequenced and compared. Further uncertainties lie in the exact relationship of soft rot pathogens to *E. coli* and *Yersinia*, suggesting the need for a wider interest in multi-tiered approaches addressing subsystem variation on case-by-case bases, and continued studies on orthology and horizontal gene transfer in enterics. Dr.

Perna and her colleaques will persist in their stoichiometric experiments and bioinformatic analyses to better understand the many "orphan reactions" discovered, and the hypothetical proteins that algorithms have located.

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